DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY,
UTTAR PRADESH, LUCKNOW

PROPOSED

EVALUATION SCHEME & SYLLABI

FOR

B. TECH. THIRD YEAR

TEXTILE TECHNOLOGY

AS PER

AICTE MODEL CURRICULUM

[Effective from the Session: 2024-25]

DR. APJ ABDUL KALAM TECHNICAL UNIVERSITY, LUCKNOW

STUDY AND EVALUATION SCHEME
B. Tech. Textile Technology

3rd Year V-SEMESTER **Effective from SESSION-2024-25**

								Evaluation Scheme					
Sl. No	**Subject Code**	**Subject Name**	**Type**	**Category**	**Periods**			**Sessional Component**		**Sessional (SW) (TS/PS)**	**End Semester Examination (ESE)**	**Total**	**Credit**
					L	**T**	**P**	**CT**	**TA**	**CT+TA**	**TE/PE**		
1	BTT501	Textile Testing-I	T	PCS	3	1	0	20	10	30	70	100	4
2	BTT502	Yarn Manufacture-III	T	PCS	3	1	0	20	10	30	70	100	4
3	BTT503	Chemical Processing of Textiles	T	PCS	3	1	0	20	10	30	70	100	4
4	BTT051/ BTT052/ BTT053	Fabric Structure, Design & Analysis/ Textile Design & Colour / Structure & Properties of Fibers	T	PEC	3	0	0	20	10	30	70	100	3
5	BTT054/ BTT055/ BTT056	Knitting Technology/Multi & Long Fibre Spinning/Textured Yarn Technology	T	PEC	3	0	0	20	10	30	70	100	3
6	BTT551	Textile Testing-I Lab	P	PCS	0	0	2	-	50	50	50	100	1
7	BTT552	Yarn Manufacturing-III Lab	P	PCS	0	0	2	-	50	50	50	100	1
8	BTT553	Chemical Processing of Textiles Lab	P	PCS	0	0	2	-	50	50	50	100	1
9	BTT 554	Internship Assessment /Mini Project/ Start-up and entrepreneurship*	P		0	0	2	-	100	100	-	100	2
10	BNC501/ BNC502	HSMC-I Audit Group/HSMC-II Audit Group	T	HSMC	2	0	0	20	10	30	70	100	-
		Total			**17**	**3**	**8**					**1000**	**23**
	Minor Degree/Honors Degree MT-2/HT-2												

*The Mini Project or internship (4 weeks) conducted during summer break after IV semester and will be assessed during V semester.

Department Elective – I
- Fabric Structure,Design & Analysis
- Textile Design & Colour
- Structure & properties of fibers

Department Elective – II
- Knitting Technology
- Multi & Long fiber spinning
- Textured yarn Technology

DR. APJ ABDUL KALAM TECHNICAL UNIVERSITY, LUCKNOW

STUDY AND EVALUATION SCHEME
B. Tech. Textile Technology

3rd Year VI-SEMESTER **Effective from SESSION-2024-25**

								Evaluation Scheme					
Sl. No	**Subject Code**	**Subject Name**	**Type**	**Category**	**Periods**			**Sessional Component**		**Sessional (SW) (TS/PS)**	**End Semester Examination (ESE)**	**Total**	**Credit**
					L	**T**	**P**	**CT**	**TA**	**CT+TA**	**TE/PE**		
1	BTT601	Textile Testing-II	T	PCS	3	1	0	20	10	30	70	100	4
2	BTT602	Advance Spinning Technology	T	PCS	3	1	0	20	10	30	70	100	4
3	BTT603	Advance Weaving Technology	T	PCS	3	1	0	20	10	30	70	100	4
4	BTT061/ BTT062/ BTT063	Technical Textiles/ Non-woven Technology/ Functional Textiles	T	PEC	3	0	0	20	10	30	70	100	3
5	BOE060A / BOE060B	Renewable Source of Energy/ Operation Research	T	OEC	3	0	0	20	10	30	70	100	3
6	BTT651	Textile Testing-II Lab	P	PCS	0	0	2	-	50	50	50	100	1
7	BTT652	Advance Spinning Technology Lab	P	PCS	0	0	2	-	50	50	50	100	1
8	BTT653	Advance Weaving Technology Lab	P	PCS	0	0	2	-	50	50	50	100	1
9	BNC601/ BNC602	HSMC-II Audit Group/HSMC-I Audit Group	P	HSMC	2	0	0	20	10	30	70	100	-
		Total			**17**	**3**	**6**					**1000**	**21**
	Minor Degree/Honors Degree MT-3/HT-4												

Department Elective – III
- Technical Textiles
- Non-woven Technology
- Functional Textiles

Open Elective-I
- Renewable Source of Energy
- Operations Research

Semester V

BTT501 Textile Testing-I (L T P 3-1-0= 4)

Course Outcome: After completing the course student will be able to:

CO1 Understand importance of textile testing and apply statistical techniques for collecting samples.
CO2 Understand the concept, significance and effect of environmental conditions on properties of textile materials.
CO3 Understand methods of measurement of various physical properties of cotton and man-made textile fibres
CO4 Understand the concept of yarn count and twist and its methods of measurement.
CO5 Understand the concept of yarn unevenness, yarn imperfections, classimat faults and hairiness and its measurement techniques.

Unit (1): Introduction of textile testing, sampling techniques, random and biased sampling, square, cut square, zoning technique, selection of sample for testing, grading of cotton fibre with respect to staple length, understanding of national and international testing standards.
Total Lectures Required = 6

Unit (2): Atmospheric conditions for testing, absolute & relative humidity, moisture regain &moisture content, importance of moisture in textiles, measurement of moisture regain & content, effect of moisture on physical & mechanical properties of textile material, factors affecting the regain, correct numerical weight, oven dry weight, Dry and wet bulb hygrometer, control of testing room atmosphere.
Total Lectures Required = 6

Unit (3): Measurement of cotton fibre properties- length, length uniformity, span length, upper-half mean length, fineness, strength, maturity, trash, colour by conventional methods and using HIV, AFIS, determination of Fibre Quality Index, Spinning Consistency Index. Measurement of manmade fibre properties- length, fineness, strength, and crimp; wrapping test for lap, sliver and roving.
Total Lectures Required = 8

Unit (4): Yarn numbering system, determination of yarn count, diameter, average & resultant count of folded yarn, relation between Ne, Denier, Tex, Nm. Instruments used for determination of yarn count, Yarn twist, classification of twist, twist measurement, direct counting method, twist-untwist method.
Total Lectures Required = 8

Unit (5): Unevenness testing of textile materials, CV% and percentage mean deviation; Optical and capacitance principles measurement; Yarn imperfections and yarn faults (Classimat system), limit irregularity, index of irregularity, Spectrogram and variance-length curve analysis; Yarn hairiness and its principle of measurement.
Total Lectures Required = 8

Grand total of lectures required = 36

Text Books & Reference Books: -

1. Principles of Textile Testing by J. E. Booth, 1961, Heywood Books, London
2. Physical Testing of Textiles by B. P. Saville, 1999, Woodhead Publishing Ltd., U. K.
3. Testing and Quality Management – Edited by V. K. Kothari, IAFL Publications, New Delhi
4. Physical Properties of Textile Fibres by J. W. S. Hearle and W. E. Morton, CRC Press, 2008.
5. NPTEL course on Textile Testing by A. Das, IIT Delhi.

BTT502 Yarn Manufacturing –III (L T P 3-1-0 =4)

Course Outcome : After completing the course student will be able to:

CO1 ; Understanding objectives and yarn formation in ring spinning and drafting theory
CO2 ; Understanding concepts of twisting and winding and package formation, spinning tension
CO3 : Understanding spinning geometry ,Ring frame efficiency, spinning faults and waste control
CO4 : Understanding compact spinning, yarn doubling techniques, twisting concepts in double yarn
CO5 : Understanding different types sewing thread, tyre cord yarn, Reeling and yarn bundling, fancy yarn manufacturing , production calculation of doubling frame, latest developments in R/F

Unit-1 : Objectives of Ring Frame, Machine design and constructional features, operations involved in ring spinning , working principle and yarn formation, Function of different elements of ring frame , objective , principle and mechanism involved in Drafting,Weighting system on ring frame, Selection of Draft
Total Lectures Required = 8

Unit-2 : Objective , principle and mechanism involved in twisting process , Constructional features of spindle , types of ring and traveler and their applications , Fibre lubrication , running on new ring ,yarn guiding elements and yarn tension control , Selection of twist , Objective , principle and mechanism involved in winding , cop structure , Building motion
Total Lectures Required =7

Unit-3 : Spinning geometry , forces acting on ring and traveler , limitation of ring spinning and Factors responsible for loss in efficiency ,Causes of end breaks and its control , Yarn faults and their remedies , classimat faults, Ring frame waste and waste collection , Calculation pertaining to draft , twist and production of ring frame.
Total lectures Required =7

Unit-4 : Compact spinning, Different methods of compaction , Advantages of compact spinning, Objectives and principle of doubling , Methods of doubling -Ring doubling , Two for one(TFO) Twisting , Concept of balanced twist in doubled yarn ,Direction of twist in doubled yarn and its relation to single yarn , Characteristics of double yarns.
Total Lectures Required =7

Unit-5 : Different types of Sewing Threads , characteristic features of sewing thread , tyre cord yarns , Objectives of Reeling , Types of reeling , construction and working of reel , Yarn bundling , Fancy yarn manufacturing, calculation pertaining to double yarn production , Latest developments in ring frame
Total Lectures Required =7

Grand Total of Lectures Required =36

Reference Books :

1. Elements of Ring frame & Doubling – Dr A.R.Khare
2. The Technology of short staple spinning –W. Klein
3. Cotton Spinning – Taggart
4. Spun Yarn Technology –Oxtoby
5. Fundamentals of spun yarn technology by C.A. Lawrence
6. Elements of Practical cotton spinning by T.K Pattabhiram
7. Spinning of man made fibres and their blends by K.R Salhotra
8. Handbook of yarn production –Lord P. R

9. Short staple Ring spinning by J Shaw
10. TFO-Technology & Techniques by HVS Murty

BTT503 Chemical Processing of Textiles (L T P 3-1-0 = 4)

Course Outcome: After completing the course student will be able to:

CO1 Understand the objectives of desizing, scouring, bleaching, heat setting and mercerization of textile materials

CO2Understanding of the processing machinery: Jigger, Winch, Jet dyeing m/c, Soft flow dyeing m/c, Pre treatment range (PTR), Mercerizing m/c etc

CO3Classification of dyes according to their mode of application

CO4Methods of printing and role & types of printing thickeners

CO5Classification of finishing and various types of finishing methods

Unit (1): Importance of textile wet processing, role of water in textile processing, dye house water standards, surfactants and their role in detergency, preparatory process; singeing, desizing, scouring, role of heat setting in synthetic material. bleaching and mercerization process of cotton goods

Total Lectures required =7

Unit (2): Brief introduction to processing machinery: Jigger and Winch, Jet and Soft flow dyeing m/c, Padding Mangle. Pretreatment range (PTR), Continuous dyeing range (CDR), Cold Pad Batch (CPB) dyeing m/c, Role of stenter and VDR m/c in finishing process. Roller printing, Rotary printing m/c and digital printing m/c.

Total Lectures required =7

Unit (3): Theory of dyeing**,** Classification of dyes according to their mode of application, Dyeing Auxiliaries, Dyeing of cellulosic material with direct, reactive, vat and sulphur dyes. Dyeing of polyester with disperse dyes, difference between exhaust and continuous dyeing, Dyeing of P/C blends

Total Lectures required =8

Unit (4): Styles and methods of printing, Role and types of printing thickeners, Properties of thickeners, Printing auxiliaries, Pigment Printing, Printing faults and their remedies.

Total Lectures required =7

Unit (5): Role of finishing process in textile processing, Introduction to finishing, Classification of finishing - Mechanical and Chemical, Various types of finishing such as Calendaring, Milling, Sanforizing, Optical brightening agent (OBA), Flame retardant and water repellent finish.

Total Lectures required =7

Total Lectures required =36

Text Books and Reference material:

1. Pre-treatment of Textile Materials for Dyeing and Printing by Prof (Dr) M. S. Parmar,NITRA Publication
2. Textile preparation and dyeing by Asim Kumar Roy Choudhary
3. Dyeing and Printing by Hemlata Jain
4. Technology of dyeing, by V.A. Shenai

5. Chemical Finishing (Garment processing Series), Published by NITRA
6. Mechanical Finishing (Garment processing Series), Published by NITRA

Departmental Elective I

BTT051 Fabric Structures, Design & Analysis (L T P 3-0-0=3)

Course Outcome: After completing the course student will be able to"
CO1 Classify various types of fabrics and understanding of basic structures
CO2 Understanding derivative weaves, various types of satin/sateen and twill weaves
CO3 Illustrate special weave structures like mock leno, honeycomb, huck-a-back and crepe weaves.
CO4 Designing of various compound structures
CO5 Understanding colour & Weave effect, Application of CAD and fabric Sett, Drafting, denting and peg plans.

Unit (1): Classification of various fabrics, representation of weave repeat. Design, draft , peg plan & Denting order. Construction of Basic weaves, plain, Twill & Satin Weaves
Total Lectures required = 7

Unit (2): derivatives of Basic weaves (rib and mat weave), ordinary twill, right hand twill, warp faced, weft faced & balanced twills. Satin regular, irregular ,Combined twills, end to end and pick-to pick combination, elongated twills, steep twills, broken twill, curved twill, Fancy twills zig-zag, herringbone , waved twill. Diaper & Diamond Designs
Total Lectures Required = 8

Unit (3): mock leno, ordinary honeycomb, brighten honeycomb, Huck-a-back and crepe weave. Derivatives of hopsack, barley corn stitched hopsack and twilled hopsack weave.
Total Lectures Required = 7

Unit (4): Principle of figuring with extra material extra warp figuring, extra weft, limitation of extra thread, Double & Triple cloth. Pile fabrics, concept of simple constructions, formation of pile; Simple terry weaves
Total Lectures Required = 8

Unit (5): colour and weave effects, construction of line effects, Hound's tooth pattern, Bird's eye and spot effects, hairline effects, step pattern, compound colour and weave effects. Fabric analysis. Application of CAD in woven fabric designs. Fabric sett, reed count, crimp and calculation of areal density;
Total Lectures Required = 6

Grand Total Lectures Required = 36

Reference Book: -

1. W. Watson Textile Design & colour Longmans Greens Co. London.
2. Z.J Grosicki Watson's Textile design and colour Newnes Buter Worth, London.
3. Z.J. Grosicki, Advance Textile Design Newnes Butter Worth, London.
4. "Nishant" A Grammar of textile.
5. Gokarnation N Fabric Structure & Design New age International New Delhi 2004

BTT052 Textile Design & Colour (L T P 3 0 0)

Course Outcome: After completing the course student will be able to"

CO1 Understanding different techniques of surface ornamentation.
CO2Understanding of Light and colour phenomena and attributes of the primary & secondary colours,
CO3 Explain Colour measurement, Primary, Secondary, Tertiary & compound colours,
CO4 Understanding about different stages of colouring of textile materials, Colour and weave effect and its classification
CO5 Unit repeating design, Geometric ornamentation, Construction of symmetrical designs, Stripe and check effect designs, artsheet work of textile design

Unit (1): History of textile design, Introduction to Woven, Printed, embroidered & painted textile designs, Textile printing techniques: direct, indirect, resist and discharge. Colour forecasting
Total Lectures required = 7

Unit (2): Light and colour phenomena, physical basis of colour, Emission & absorption theory of light, Colour vision and light theory of colours, Complementary colours, Chromatic circle, Pigment theory of colours, Brewster circle, Attributes of the primary & secondary colours
Total Lectures required = 7

Unit (3): Colour measurement, Primary, Secondary, Tertiary & compound colours, Biren's triangle, Modification of colours, Coloured greys, Colours in combination, Colour contrast, contrast in hue, contrast of tone, colour harmony, Relative spaces occupied by colours, divisional colours, Application of colours, Mixed colour effect,.
Total Lectures required =8

Unit (4): Composition of designs, Condition to be observed during ornamentation of fabrics, Mode shade, Harmony of succession, gradation of hue, Different stages of colouring of textile materials, Bases of Textile design, One third and one fourth drop design, Half drop and drop reverse design.
Total Lecture required = 8

Unit (5): Unit repeating design, Geometric ornamentation, Construction of symmetrical designs, Stripe and check effect designs, Sari border / vertical border design, Factor affecting the woven designs, reversing inclined figure, Diamond, Ogee, & diagonal waved line base, applications of colours. Art sheet based question covering all above units
Total Lecture required = 6

Grand total Lectures required = 36

Reference Books:
1. W. Watson- Textile design and colour
2. Traditional Textile designs B K Behera

BTT053 Structure & Properties of Fibres (L T P 3 0 0)

Course Outcome: After completing the course student will be able to:

CO1 : Describe various models of physical structure of fibre and Analyze structure–property relationship
CO2 : Explain various properties of fibres.
CO3 : Characterize and measure various physical and chemical structure of fibres by using instrumental techniques.
CO4 : Derive various mathematical models of fibre structures and properties.
CO5 : Determine the effects of different factors influencing fibre properties.

Unit (1): Basic structural features of fibre, Structure of Cotton, wool, silk, and other textile fibres, relation between fibre structure and fibre, Methods of estimating molecular weight, orientation, crystallinity & crystalline orientation of fibre forming polymer, Overall orientation by sonic modulus tester.
Total Lectures Required = 7

Unit (2): Principle of microscopy , Microtomy , Preparation and mounting of specimen for longitudinal view and cross sectional view**,** Concept of scanning electron microscope (SEM)**,** Concept of transmission electron ,microscope (TEM) , Fourier Transform Infrared Spectroscopy (FTIR), Atomic force microscopy .
Total Lectures Required = 7

Unit (3): Thermal behaviour of textile fibres by Differential Scanning Calorimeter (DSC), Thermal gravimetric analysis (TGA), Thermo mechanical Analyser (TMA) Density gradient column, Preparation of density gradient column, Crystallinity by density gradient column.
Total Lectures Required = 7

Unit (4): Optical properties of fibres (2), Birefringence behaviour, dielectric properties, fibre Friction-fibre to fibre friction , fibre to metal friction , fibre friction measurement, Significance of fibre friction property , fibre fracture.
Total Lectures Required = 7

Unit (5): Creep behaviour, concept of moisture absorption by fibres, Moisture absorption, heat of absorption, differential heat of absorption, integral heat of absorption, Quantitative theory of heat moisture absorption, Rate of moisture absorption. Application of heat and moisture absorption properties of fibre .
Total Lectures Required = 8

Grand Total of lectures required = 36

Reference Book: -

1. Manufactured fibre technology by V.B. Gupta, V.K. Kothari
2. Physical properties of fibre by J.W.S. Hearle
3. Thermal behaviour of material by Turi
4. Modern yarn production by Ray
5. ASTM Standard books

Departmental Elective II

BTT054 Knitting Technology (L T P 3 0 0)

Course outcome: After completing the course student will be able to:

CO1	Difference between knits and woven, knitting terms, elements and their definitions, comparison of warp and weft knitting
CO2	Elements of knitting, Classification of knit structures, loop formation on: single jersey, rib and interlock machine
CO3	Flat bed knitting machine, Loop length geometry, science of knitting and their numerical problems
CO4	Warp knitting machines, classification of warp knitting, Guide bars construction
CO5	Modern developments in weft and warp knitting, spacer fabrics, numerical calculations

Unit 1: Basic knitting terms and definitions (course, wale, CPI, WPI, stitch density, loop length), Comparison of weaving and knitting, comparison of warp and weft knitting, Preparation of yarns for weft and warp knitting; Classification of weft knitting machines; Structural Elements of Weft Knitting: Needle loop, sinker loop, technical face, technical back, knit stitch, tuck stitch, purl stitch, miss stitch, needle numbering system

Total lectures required=6

Unit 2: Mechanical elements of weft knitting: needles, and their types (latch needle, bearded needle and compound needle), sinkers, cams, cylinder, jack; knitting action of all needles; Knitting cycle and yarn path of weft knitting; Four basic weft knitting structures- single jersey, rib, purl and interlock (their structures, loop formation, machine construction, fabric characteristics, use and derivatives), Circular weft knitting machines

Total lectures required=8

Unit 3: Flat bed knitting machine: Process of loop formation, cam track, features, and structures produced; Relation between yarn count and machine gauge; Science of Knitting: Concept of loop length, loop length geometry, knitting tension, fabric relaxation and shrinkage, tightness factor, robbing back, spirality, all related numerical

Total lectures required=7

Unit 4: Warp knitting principles: open loop, closed loop, overlap and under lap; Construction of warp knitted fabrics, principle, operation and comparison of Tricot and Raschel machines, mechanism of needle and guide bar movements, Single needle bar structures; Multiple guide bar construction: lock knit, reverse lock knit, satin, sharkskin structures and properties; Swinging and shogging of guide bars

Total lectures required=8

Unit 5: Modern developments in weft and warp knitting machines; calculations regarding production, GSM, stitch density etc of weft and warp knitting; spacer knitted fabrics; causes and remedies of faults in knitted fabrics

Total lectures required=7

Grand total of lectures required=36

References and Textbooks-

1. Knitting technology- Chamberlin
2. Knitting technology-W.J. Spencer
3. Knitting calculation-Chamberlin
4. Weft knitting Vol. 1&2- Published by IIT New Delhi
5. Knitting- NCUTE

BTT055 Multi & Long Fibre Spinning (L T P 3-0-0= 3)

Course Outcome: After completing the course student will be able to"

CO1 Characteristics of manmade fibres, Spinnability, blending, and its objectives,
CO2 Spinning of dyed fibres, estimation of blends intimacy, factors affecting the blend irregularity,
CO3 Production of bulked yarn, characteristic difference in the physical & mechanical properties,
CO4 Retting of flax, & jute, Jute & flax spinning,
CO5 Woollen, semi-worsted & worsted systems of spinning

Unit 1: Characteristics of manmade fibres, Spinnability, blending, and its objectives, Spinnability, blending & its objectives, processing of Man- made fibres & blends on short, medium and long staple spinning systems.
Total Lectures Required = 8

Unit 2: Spinning of dyed fibres, estimation of blends intimacy, factors affecting the blend irregularity, structural properties of blended yarns.
Total Lectures Required = 8

Unit 3: Production of bulked yarn, characteristic difference in the physical & mechanical properties of various long staple fibres & their influence in the choice of preparatory & spinning machinery.
Total Lectures Required = 8

Unit 4: Retting of flax, & jute, Jute & flax spinning, manufacturing of spun silk.
Total Lectures Required = 6

Unit 5: Woollen, semi-worsted & worsted systems of spinning.

Total Lectures Required = 6

Grand total lectures required =36

Reference book:-
1. Wool Spinning by Vickerman, Abhishek Publication
2. Principle of woolen spinning by Priestman
3. Woolen & Worsted yarn manufacture by J. W Redcliff
4. Jute Spinning Calculation by Andrew Smith
5. Worsted Drawing & Spinning by Miles

BTT056 Textured Yarn Technology (3-0-0=3)

Course Outcome: After completing the course student will be able to"

CO1 Importance of texturising, Methods of texturising
CO2 Structural geometry of textured yarn, characterization of textured yarn
CO3 Textured yarn properties and fabric characteristics, principles of draw texturing
CO4 Principles of air bulking and properties of air bulked yarn,
CO5 Morphological chances induces by mechano-thermal forms of texturing

Unit (1): Importance of texturising, Methods of texturising, Basic principles of heat setting and texturising, False twisting, Process variables, developments of false twist texturing machines.
Total Lectures Required = 7

Unit (2): Structural geometry of textured yarn, characterization of textured yarn, Stuffer box crimping, Edge crimping & other methods of texturing thermoplastic yarns.
Total Lectures Required = 7

Unit (3): Textured yarn properties and fabric characteristics, principles of draw texturing, Draw texturising machines, and properties of draw textured yarn, Effect of process variables, time, temperature, twist and tension on properties of textured yarn.
Total Lectures Required = 8

Unit (4): Principles of air bulking and properties of air bulked yarn, Texturing of non thermoplastic yarns cross linking and effect of process variables on properties of textured yarn, Texturing with the aid of solvent.
Total Lectures Required = 7

Unit (5): Yarn characteristics, Morphological chances induces by mechano-thermal forms of texturing and their effect on properties of textured yarns.
Total Lectures Required = 7

Grand total lectures required =36

Reference book:-

1. Textured yarn Technology by J.W.S.Hearle
2. Synthetic fibre by Vaidhya

5th Semester Lab Syllabus

BTT551 Textile Testing-I: Laboratory Experiments (L T P 0 0 2)

1. Determination of various cotton fibre properties by using HVI/conventional methods
2. Determination of Trash% in cotton fibre by Trash Analyzer
3. Determination of cotton fibre maturity by various methods
4. Determination of various man-made fibre properties
5. Determination of count and count strength product (CSP) of spun yarns
6. Determination of twist per unit length of single and plied yarns
7. Determination of Unevenness, Imperfections and hairiness of yarn
8. Measurement of moisture regain and moisture content

BTT552 Yarn Manufacturing-III (L T P 0 0 2)

1. To study the objectives, working principle, material flow and advantages / disadvantages of ring spinning.
2. To study the various parts of ring frame machine and their functioning.
3. To study the drafting system of ring frame and calculate back draft , main draft and total draft assuming appropriate parameters.
4. To study the various package defects at ring frame with their possible reasons.
5. To study on the gearing diagram and calculate twist per inch and twist constant in the ring frame .
6. To calculate the front roller delivery speed & production of ring frame machine assuming appropriate process parameters.
7. To study the ring doubling machine and learn how to calculate production / spindle.
8. To make neat sketch of reeling machine and find swift circumference and production / spindle.
9. To study the various parts of Two for one Twister (TFO) and their function.
10. To study the various settings and speed of moving parts of Two for one Twister (TFO) and calculate the production per spindle per hour in Kg

BTT553 Chemical Processing of Textiles (L T P 0 0 2)

1. Learn how to carry out desizing, scouring and bleaching of cotton goods
2. Learn how to carry out dyeing of cotton goods using direct dyes.
3. Learn how to carry out dyeing of cotton goods using reactive dyes.
4. Learn how to carry out dyeing of cotton goods using vat dyes.
5. Learn how to carry out dyeing of polyester-cotton (P/C) blends.
6. Learn how to carry out printing of cotton materials using direct, resist, and discharge methods.
7. Learn how to carry out application of water repellent finish and optical brightening agent on cotton materials.
8. Learn how to determine colour fastness to washing.
9. Learn how to determine colour fastness to rubbing.
10. Learn how to determine colour fastness to light

Semester VI

BTT601: Textile Testing-II (L T P 3-1-0= 4)

Course Outcome: After completing the course student will be able to:

CO1: Develop understanding of various principles of textile testing, modulus and yield point and tensile testing of yarns.
CO2: Develop understanding of various testing instruments used in tensile testing of yarn and fabrics
CO3: Understand various physical properties of fabrics, their method of testing and instruments involved in their measurement.
CO4 Understand various chemical properties of fabrics, their method of testing and instruments involved in their measurement.
CO5: Develop understanding of testing of technical textiles and non woven materials.

Unit (1): Tensile testing: CRE, CRL and CRT principles, stress-strain curve, various methods for finding of yield point, methods for finding of various modulus, tensile testing of single yarn and lea, count-strength product; Effect of gauge length, rate of extension and relative humidity on tensile properties, weak-link theory.
Total Lectures Required = 7

Unit (2): Knowledge about various yarn and fabric strength testing instruments- single yarn strength tester, lea strength tester, fabric tensile strength tester, bursting strength tester, tear strength tester.
Total Lectures Required = 7

Unit(3): Knowledge of various physical properties of fabric and equipment used- thread density of woven and knitted fabrics, crimp, weight, cover factor, abrasion resistance, pilling, air permeability, thickness, drape, stiffness, crease recovery, wrinkle recovery. Introduction to Kawabata Evaluation System, Fabric Assurance by Simple Testing,
Total Lectures required =8

Unit (4): Determination of composition of various blends, shrinkage, pH, concept of color theory, measurement of color fastness to light, rubbing, washing, sublimation, perspiration, drycleaning and sea water; thermal resistance and transmission, water repellency
Total Lectures required =8

Unit (5): Introduction to testing of technical textiles (filters, geo-textiles, bullet-proof fabrics, fire-protective fabrics etc.); Testing of antibacterial fabrics. Testing of nonwoven materials
Total Lectures required =6

Grand total of lectures required = 36

Text Books & Reference Books: -

1. Principles of Textile Testing by J. E. Booth, 1961, Heywood Books, London
2. Physical Testing of Textiles by B. P. Saville, 1999, Woodhead Publishing Ltd., U. K.
3. Testing and Quality Management – Edited by V. K. Kothari, IAFL Publications, New Delhi
4. Physical Properties of Textile Fibres by J. W. S. Hearle and W. E. Morton, CRC Press, 2008.
5. NPTEL course on Textile Testing by A. Das, IIT Delhi.

BTT602 Advance Spinning Technology (L T P 3 1 0=4)

Course Outcome: After completing the course student will be able to-

CO1 : Understanding ring spinning limitations, principle of OE spinning , Classification of new spinning technologies
CO2 :Understanding principle & yarn formation of rotor spg ,yarn structure & properties of rotor yarn
CO3 : Understanding concept of friction spinning , Dref-2 and Dref-3 machines and yarn formation, yarn structure , properties and application of friction yarns
CO4 : Understanding principle and yarn formation of air jet spinning , vortex spinning, limitations and uses of air spinning technology , characteristics of air jet and vortex yarns
CO5 :Understanding concepts of self twist and twist spinning, Twist-less spinning , electrostatic spinning , wrap spinning, core spinning and felting process .

Unit-1 : Limitations of ring spinning , causes leading to advent of non-conventional system of spinning ,Principle of unconventional method of yarn manufacturing open-end spinning process , Advantages and limitations of open-end spinning process , Yarn structure and end use applications of open end yarns , Classification of new yarn spinning technologies. Brief description of new yarn spinning technologies.
Total Lectures Required =7

Unit-2 : Rotor spinning –Objectives and working principle of rotor spinning , Operations involved in rotor spinning , Design features of rotor spinning machine and yarn formation, Design features of opening unit , rotor , naval and yarn withdrawal tube, Raw material requirements, Yarn structure, Yarn characteristics-comparison with ring yarn , Latest development in Rotor spinning
Total Lectures Required =7

Unit-3 : Friction spinning – principle ,Dref -2 and Dref-3 spinning , Design features of Dref-2 and Dref-3 machines , working principle and yarn formation in Dref-2 and Dref-3 , Raw material requirements , Limitations of friction spinning , Yarn structure and Properties of friction spun yarn, End use applications of friction yarns
Total Lectures Required =7

Unit-4 : Air jet spinning-Principle , concept of false twist , Double nozzle air jet spinning- working principle and yarn formation , Vortex spinning – working principle and yarn formation , Raw material requirements , effect of process variables on yarn formation , Yarn characteristics of air jet spun yarns. Limitations of air jet yarns ,End use applications of air jet spun yarns.
Total Lectures Required =8

Unit-5 : Self twist spinning – Rubbing Technique , Twist spinning- Plyfil spinning and SIRO spinning , Twist less spinning – TWILLO process , Bobtex ICS process , Wrap spinning- parafil spinning , Electrostatic spinning , Core spinning , Felting Process – Periloc yarn
Total Lectures Required = 7

Grand Total of Lectures Required = 36

Reference Books :

1. New Spinning Technology Vol 4 & 5 – W. Klein
2. Fundamentals of spun yarn technology By C. A Lawrence
3. Spun Yarn Technology – Oxtoby
4. New Spinning Systems –Mahendra Gowda R V (NCUTE Publication ,IIT Delhi ,2006)

BTT603 Advance Weaving Technology (L T P 3 1 0=4)

Course Outcome: After completing the course student will be able to:

CO1 Understand different principles of Shuttle-less weaving,
CO2 Understand the mechanism of projectile weaving, different weft feeding mechanism,
CO3 Understand different rapier weaving mechanism, importance of control,
CO4 Understand of principle of Air-Jet weaving, mechanism of beat-up in air-jet weaving
CO5 Understand water-jet weaving mechanism, uses of water-jet loom, machine parameters for water-jet loom.

Unit 1: Principle of operation of shuttle less loom, Classification of shuttle less looms, Comparative study of various systems of weft inserting, advantage of shuttle less looms over shuttle looms.
Total of lectures required=7

Unit-2: Basic operational principle of projectile weaving machinery, types of projectile, projectile guides; Torsion bar picking mechanism of Sulzer projectile loom, beating, Selvedge formation, multi-colour weft feeding on projectile loom. sequence and timing of operations
Total of lectures required=7

Unit-3: The Rapier system of weft insertion, Classification of Rapier looms and its description, Rapier drives, working of some standard drive mechanisms. flexible and rigid rapier, weft transfer & weft Control mechanism, loom timing , rapier design, tip & loop transfer mechanism of rapier loom. Weft insertion rate & production calculation.
Total lectures required=7

Unit-4: Working principle of Air-jet weaving, essential requirements of air-jet loom, weft measuring, weft tensioning devices, beating-up, weft gripping, weft cutting, weft stop motion selvedge formation, Tendom nozzle and relay nozzles, quality of compressed air for air jet loom, multiphase weaving
Total lectures required=8

Unit-5: Weft insertions of water-jet loom , general description of the water-jet loom, weft supply system, tension measuring device, water stream for weft insertion, fundamental problems of hydraulic weft insertion, Electronic dobby and electronic jacquard.
Total lectures required=7

Grand total of lectures required=36

Reference books

1.Modern development in weaving machine by V wray Duxbury
2. Weaving mechanism, machine and Management by Talukdar. Sriramulu, Mahajan Publishers,
3. Weaving Mechanism by Robinson.
4. Modern weaving machinery Ormerd.
5. Principles of Woven Fabric Manufacturing by A. Majumdar, CRC Press (2016).
6. Principles of Weaving R. Marks & A. T. C. Robinson

Department Elective III

BTT061 Technical Textiles (L T P 3 0 0 = 3)

Course Outcome: After completing the course student will be able to:

CO1 Understand technical textiles, industrial textiles manufacturing methods, products, filter fabrics and its mechanisms, finishing treatments, and nanotechnology.

CO2 Protective textiles products and its manufacturing, heat transfer mechanism, bullet proof and camouflage fabrics.

CO3 Medical textile categories, applications of medical textiles products, testing methods, basics of sports textiles.

CO4 Products and manufacturing techniques of smart textiles, automotive textiles, and agro textiles.

CO5 Classifications and manufacturing of geo textiles, testing of geo textiles, and overview of composites.

Unit 1: Introduction to technical textile, Classifications of technical textiles, Industrial textile – filtration mechanisms, filter fabric constructions- woven, needle felt & knitted filter fabric, finishing treatment of filter fabric, essential requirements of good filter fabric, hoses, ropes and nets, tarpaulins. Nanotechnology – methods to manufacture nanofibers, advantages.

Total Lecture required = 8

Unit 2: Overview of protective textiles, Manufacture and properties of protective textiles- water proof/coated and water repellent, antimicrobial, flame retardant – conduction, convection, and radiation heat transfer, Nuclear biological & chemical (NBC) resistance, mechanical resistance - bullet proof, cut proof, stab proof. Camouflage concealment and deception.

Total Lecture required = 7

Unit 3: Medical textiles, fibres used, classification of medical textiles, manufacturing, functions and testing of non-implantable products, Implantable products, Extra-corporal devices, Healthcare/ hygiene products. Sports textiles – sportswear, sports goods, hot air balloon.

Total Lecture required = 7

Unit 4: Smart textiles, brief introduction of smart textiles, classification of smart textiles, passive smart textiles, active smart textiles, brief discussion of smart shirt, smart suit, musical jacket, space suit. Automotive textiles: tyre cord, seat belt, air bag, interiors and exteriors trim. Agro textiles: Shade net, green house film, Mulch net, crop cover, anti-hail and bird protection net, finishing net etc.

Total Lecture required = 8

Unit 5: Introduction of geo textile, classification of geo textiles, functions of geo textile, objective of geo textiles, manufacturing of geo textile, essential properties of geo textiles- Mechanical determinants, Hydraulic determinants, durability determinants. Composites – Introduction, types of textile reinforcement structures, manufacturing, and applications.

Total Lecture required = 6

Grand total of lectures required = 36

Reference books:

1. Hand Book of Technical Textiles – A. R. Horrocks & S. C. Anand
2. Wellington Sears Handbook of Industrial Textiles – Sabit Adanur
3. Medical and Healthcare Textiles – S. C. Anand
4. Nanotechnology in Textiles: Theory and Applications – Rajesh Mishra and Jiri Militky
5. Smart Fibres, Fabrics and Clothing – Xiaoming Tao

BTT062 Non woven Technology (L T P 3 0 0=3)

Course Outcome: After completing the course student will be able to:

CO1 Concept about felts and nonwoven, Classification of nonwoven fabrics
CO2 Various method of web formation, web characteristics and their influence on properties of nonwoven fabrics, needle punching manufacturing,
CO3 Nonwoven fabric by hydro entanglement, Description of hydro entanglement machine, other methods to prepare nonwovens,
CO4 Nonwoven fabric by Stitch bonding, Nonwoven fabric by chemical bonding
CO5 Knowledge about latest development in non-woven industry.

Unit 1: National and international scenario on non-woven fabric production, Definition as per INDA and EDANA, Concept about felts and non-woven, Classification of non-woven fabrics, fibres for non-woven fabrics, Felt manufacturing process.
Total Lectures Required =7

Unit 2: Classification of web laying and web bonding systems, Comparison of woven, knitted and nonwoven. Various method of web formation, web characteristics and their influence on properties of non-woven fabrics, Nonwoven fabric by Needle punch, Description of needle punching machine, effect of process variables on properties of needle punch fabric, applications.
Total Lectures Required =8

Unit 3: Non-woven fabric by hydro entanglement, Description of hydro entanglement machine, effect of process variables on properties of hydro entanglement non-woven fabric, Nonwoven fabric by adhesive bonding, mechanical bonding, Melt blown process of non-woven fabric manufacturing, Electrospinning of nonwoven process manufacturing and advantages, applications of hydro entanglement, adhesive bond, mechanical bonded, melt blown and electrospinning nonwovens.
Total Lectures Required =8

Unit 4: Non-woven fabric by Stitch bonding, Nonwoven fabric by chemical bonding, Nonwoven fabric by bonding with thermoplastic adhesives, Nonwoven fabric by Spun laced, Effect of process variables on properties of stitch bonding, chemical bonding spun laced non-woven fabrics, STRUTO vertical lap nonwoven process and parameters.
Total Lectures Required =7

Unit 5: Flocked fabric, Laminates, latest development in non-woven industry: ultrasonic bonding, Infra-red bonding, bonding by bi-component fibres. Application of various nonwoven fabrics.
Total Lectures Required =6

Grand total of Lectures required= 36

Reference & Text Books

1. Nonwoven – N.N. Banerjee
2. Nonwovens: process, structure, properties and applications by T. Karthik, Prabha Karan C., R. Rathinamoorthy – WPI publishing, 2016.
3. Hand Book of Nonwoven – Russell S J (WPI publications- 2nd Edition 2024)
4. Hand Book of Technical Textiles – A. R. Horrocks & S. C. Anand
5. Nanotechnology in Textiles: Theory and Applications – Rajesh Mishra and Jiri Militky

KTT 063 Functional Textiles (L T P 3 0 0=3)

Course Outcome: After completing the course student will be able to:

CO1 Classification of functional clothing, Functional finishes, etc
CO2 Nuclear biological Chemical protective clothing, extreme cold clothing; design, mechanism and applications,
CO3 Medical functional clothing: therapeutic and rehabilitative clothing,
CO4 Sportswear clothing and moisture management in sportswear,
CO5 Smart Textiles, classification of smart textiles, Intelligent textiles, mechanism of various types of smart textiles.

Unit 1: Definition of functional clothing, Classification of functional clothing, Functional finishes, ways to apply functional finishes on textile surfaces, Functional clothing market review.
Total Lectures Required =6

Unit 2: Functional Protective clothing: Nuclear biological Chemical protective clothing, extreme cold clothing; design, mechanism and applications, Fire retardant clothing, approaches to achieve fire retardancy in clothing, mechanism of fire retardancy, Soft and hard body armour , super thickening fluids for body armour, water proof breathable fabrics, ways to achieve waterproof textiles and breathability in textiles.
Total Lectures Required =7

Unit 3: Medical functional clothing: therapeutic and rehabilitative clothing, bio sensing clothing, wound healing promoting dressings, antimicrobial sutures.
Total Lectures Required =7

Unit 4: Sportswear clothing: moisture management in sportswear, compression textiles, Aerodynamics, Spacesuit design and development, woven and knitted sportswear.
Total Lectures Required =7

Unit 5: Cosmetic-textiles: definition, classification, mechanism to develop various cosmetic effects in textiles, various cosmetoin gradients, worldwide scene of cosmeto-textiles, Smart Textiles, classification of smart textiles, Intelligent textiles, mechanism of various types of smart textiles, Wearable electronics.
Total Lectures Required =9

Grand total of lectures required = 36

Reference Books:
1. Functional Finishes for Textiles: Improving Comfort, Performance and Protection
2. Functional Textiles and Clothing, G. Thilagavathi., M. Parthiban, S. Viju
3. Electronics in Textiles and Clothing: Design, Products and Applications L. Ashok Kumar, C. Vigneswaran, CRC Press
4. Smart Clothes and Wearable Technology, J. McCann (Editor), David Bryson Woodhead Publishing Series in Textiles

6th Semester Lab Syllabus

BTT651 Textile Testing-II: (L T P 0 0 2)

1. Determination of thread density and crimp in woven and knitted fabric
2. Determination of weight of a fabric
3. Determination of tensile strength of a woven fabric
4. Determination of fabric tear strength
5. Determination of fabric bursting strength
6. Determination of abrasion resistance of fabric
7. Determination of pilling resistance of fabric
8. Determination of air permeability of fabrics
9. Determination of crease recovery of fabrics
10. Determination of stiffness of fabric
11. Determination of P/C and P/V blend compositions
12. Determination of fabric shrinkage
13. Determination of FR properties of protective textiles

BTT652 Advance Spinning Technology (L T P 0 0 2)

1. To study the working principle of Open End Spinning and various yarn spinning technologies based on Open End Spinning.
2. To study the material Flow through and yarn formation in rotor spinning .
3. To study the function and role of different parts of rotor spinning.
4. To study the operating Principle and material flow of DREF-2.
5. To study the operating Principle of Air jet spinning technology with neat sketch .
6. To study the operating principle of Vortex Spinning with passage of material through machine .
7. To study the operating principle of Electrostatic Spinning with passage of material through the machine.
8. To study the operating principle of Wrap spinning with passage of material through the machine .
9. To study the yarn formation and passage of the material through SIRO spinning .
10. To study the principle and yarn formation in Bobtex ICS twist less spinning Process .

BTT653 Advance Weaving Technology (L T P 0 0 2)

1. To study picking mechanism of Sulzer projectile loom
2. To study picking mechanism of Rapier system by Gabler mechanism
3. To study picking mechanism of Rapier system by Dewas mechanism
4. To study picking mechanism of air-jet loom
5. To study picking mechanism of water-jet loom
6. To study electronic dobby in modern loom
7. To study picking of electronic jacquard in modern loom
8. To study of selvage formation technique of various modern loom